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COMP3270
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HW4

1. Breath-First Search

Vertex	d	π
1	∞	NIL
2	∞	NIL
3	0	NIL
4	∞	NIL
5	∞	NIL
6	1	3

2. Depth First Search

Vertex	d	f
u	1	12
v	2	11
w	3	10
x	4	9
y	5	8
z	6	7

Tree edges: (u,v), (v,w), (w,x), (x,y), (y,z)
Forward edges: (u,w), (u,x), (v,x), (v,y)
Cross edges**:** (w,y), (x,z)
Back edges: None

3. Topological Sort

From decreasing finishes times: u → v → w → x → y → z

4. CLRS 22.1-1

Adjacency list representation: **O(V + E)**
Adjacency matrix representation: **O(V^2)**
Transpose graph: obtained by reversing all edges

5. CLRS 22.2-1 (Shortest paths from s)

Distances
d(s)= 0

d(t)= 3
d(y)= 5
d(x)= 9
d(z)= 11
Predecessors
π(s)= NIL
π(t)= s
π(y)= s
π(x)= t or y
π(z)= y

6. CLRS 22.3-1

(a) Bellman-Ford (source = z)

Final distances
d(z)= 0
d(s)= 2
d(t)= 2
d(x)= 4
d(y)= 9
Predecessors
π(s)= z
π(t)= x
π(x)= y
π(y)= s
π(z)= NIL

(b) Modify (z,x)= 4, source = s

Final distances
d(s)= 0
d(t)= 2
d(x)= 4
d(y)= 7
d(z)= -2
Predecessors
π(s)= NIL
π(t)= x
π(x)= y
π(y)= s
π(z)= t

7. Updating MST

Strategy

- Add new vertex: v
- Examine all edges: (v,u)
- Add the minimum-weight edge connecting (v) to the MST

Complexity**:** $O(V)$

8. K-Means (1 epoch)

Points:

A1(2,10), A2(2,5), A3(8,4), A4(5,8),
A5(7,5), A6(6,4), A7(1,2), A8(4,9)

Initial centers:

A1, A4, A7

(a) Clusters

C1: {A1, A8}
C2: {A3, A4, A5, A6}
C3: {A2, A7}

(b) New Centroids

C1=(3, 9.5)
C2=(6.5, 5.25)
C3=(1.5, 3.5)

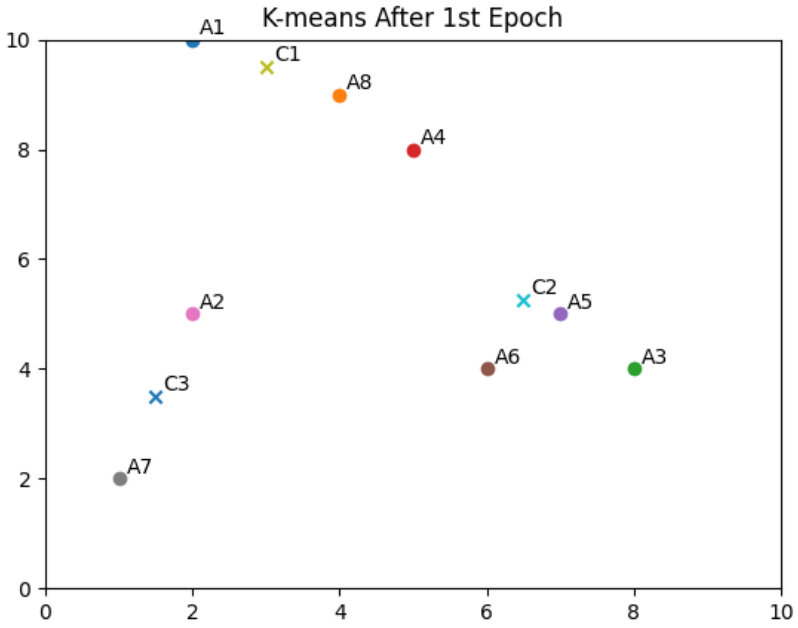
(c) First Epoch

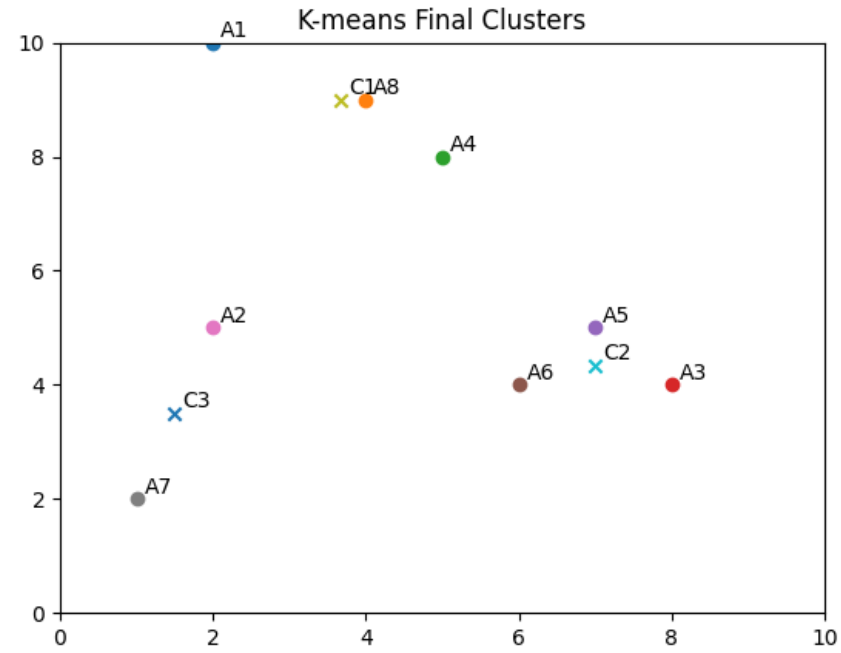
Points grouped into:

- C1: A1, A8
- C2: A3, A4, A5, A6
- C3: A2, A7
  Centroids marked with **X**

(d) Final Convergence

Clusters stabilize as:
C1: A1, A4, A8
C2: A3, A5, A6
C3: A2, A7
(Graphs are on the next page)





Clusters updated based on Euclidean distance

Centroids recomputed as mean of assigned points

Algorithm converges after ~2 additional iterations